

Mathematical Formalization of LUNA: A Lunar-Inspired Metaheuristic Optimization Algorithm

*Theoretical Foundation for Systematic Literature Review
and Benchmark Paper*

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Abstract

This document provides a rigorous mathematical formalization of the LUNA (Lunar-inspired Update & Navigation Algorithm), a population-based metaheuristic optimization algorithm inspired by the lunar cycle and tidal forces. The formalization covers five core algorithmic components—population initialization, phase-angle dynamics, exploration via Gaussian random walk, exploitation via simplified gravitational pull, and greedy selection—and presents the complete pseudocode, parameter specification, and a full computational complexity analysis demonstrating that LUNA runs in $\mathcal{O}(T \cdot N \cdot (D + C_f))$ time and $\mathcal{O}(N \cdot D)$ space, on par with established metaheuristics such as PSO, GA, DE, GSA, and WOA. A side-by-side mathematical comparison with these five algorithms reveals that LUNA’s exploration operator is a special case of (μ, λ) -Evolution Strategy mutation, while its exploitation operator is a mass-normalized variant of the Gravitational Search Algorithm force law; the phase-switching mechanism is conceptually parallel to the hard transition used in WOA. Based on this dissection, we identify three concrete novelty gaps that currently limit LUNA’s theoretical contribution relative to existing algorithms: (G1) static parameter problem, (G2) hard phase transition, and (G3) absence of personal-best memory. Each gap is paired with a formalized v2 enhancement. The document concludes by mapping how this formalization serves as the theoretical foundation for an upcoming Systematic Literature Review (SLR) and the benchmark paper for LUNA v2.

Keywords: metaheuristic optimization, lunar-inspired algorithm, swarm intelligence, mathematical formalization, exploration–exploitation trade-off, computational complexity.

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1 Introduction

LUNA (Lunar-inspired Update & Navigation Algorithm) is a population-based metaheuristic optimization algorithm whose high-level inspiration is the lunar cycle and the associated tidal forces exerted on terrestrial bodies. The algorithm partitions its iterative budget into two phases—an *exploration phase* and an *exploitation phase*—whose interpretation loosely parallels the waxing and waning of the moon. The current prototype (hereafter *LUNA v1*) was released as open-source code and preliminarily applied to electric vehicle charging station (EVCS) placement in small cities, and to short-term load forecasting via a hybrid LUNA–ANN scheme for an Indonesian regional utility.

The aim of this document is to provide a *rigorous, reproducible* mathematical formalization of LUNA v1 that meets the notational standards expected by journals such as *Swarm and Evolutionary Computation*, *Applied Soft Computing*, and *Information Sciences*. Such a formalization is a prerequisite for two subsequent research deliverables. First, it provides the precise vocabulary needed to construct a Systematic Literature Review (SLR) on physics- and astronomy-inspired metaheuristics, in which LUNA must be taxonomically positioned relative to its true mechanistic cousins (GSA, MVO, BHA, TFA, BB-BC, CFO, GbSA). Second, it provides the reference specification against which LUNA’s empirical performance must be benchmarked, and against which a redesigned LUNA v2 must be evaluated.

The structure of this document is as follows. Section 2 fixes the notational conventions used throughout. Section 3 states the underlying optimization problem. Section 4 formalizes the five algorithmic components of LUNA v1. Section 5 presents the complete pseudocode. Sections 6 and 7 provide the parameter specification and the complexity analysis, respectively. Section 8 offers a side-by-side mathematical comparison with five established algorithms. Section 9 discusses convergence properties. Section 10 identifies three novelty gaps and proposes formalized v2 enhancements. Section 11 closes with the next research deliverables.

2 Notational Conventions

Throughout this document we adopt the following conventions. Scalars are denoted by italic lowercase letters (x, t, α). Vectors are denoted by bold lowercase letters ($\mathbf{x}, \mathbf{x}_i$). Sets are denoted by calligraphic letters ($\mathcal{X}, \mathcal{D}$). The Euclidean norm is $\|\cdot\|$. Random variables drawn from a distribution $\mathcal{D}$ are written as $X \sim \mathcal{D}$. Table 1 summarizes the principal symbols.

3 Problem Formulation

LUNA is designed to solve continuous, box-constrained optimization problems of the form

$$\mathbf{x}^* = \arg \min_{\mathbf{x} \in \Omega} f(\mathbf{x}), \quad \Omega = \bigtimes_{j=1}^D [L_j, U_j] \subset \mathbb{R}^D, \quad (1)$$

where $f : \mathbb{R}^D \rightarrow \mathbb{R}$ is the (possibly non-convex, non-differentiable) objective function, D is the problem dimension, and $[L_j, U_j]$ defines the search interval along the j -th coordinate. No assumption is made on the smoothness, convexity, or unimodality of f . In the LUNA v1 prototype, the canonical setting is $[L_j, U_j] = [-10, 10]$ for all j ; this is a tuning choice, not a structural constraint.

Definition 3.1 (Feasible Region). The feasible region $\Omega \subset \mathbb{R}^D$ is the closed Cartesian product of per-dimension intervals $\bigtimes_{j=1}^D [L_j, U_j]$. A candidate solution $\mathbf{x}$ is feasible if and only if $\mathbf{x} \in \Omega$.

Definition 3.2 (Global Optimum). A point $\mathbf{x}^* \in \Omega$ is a global optimum of f over Ω if and only if $f(\mathbf{x}^*) \leq f(\mathbf{x})$ for all $\mathbf{x} \in \Omega$.

Table 1: Principal notation used throughout the document.

Symbol	Meaning	Type
T	Maximum number of iterations (terminal time)	integer
t	Current iteration index, $t \in \{0, 1, \dots, T-1\}$	integer
N	Population size	integer
D	Problem dimension	integer
$\mathbf{x}_i^{(t)}$	Position vector of the i -th individual at iteration t	$\mathbb{R}^D$
$\mathbf{x}_{\text{best}}^{(t)}$	Best-so-far position up to iteration t	$\mathbb{R}^D$
$f(\cdot)$	Objective (fitness) function, $f : \mathbb{R}^D \rightarrow \mathbb{R}$	function
$\mathcal{F}_i^{(t)}$	Fitness value of individual i at iteration t	scalar
G	Gravitational constant (LUNA v1: static, $G = 2.5$)	scalar
σ	Standard deviation of Gaussian random walk (LUNA v1: $\sigma = 0.5$)	scalar
$\alpha^{(t)}$	Phase angle at iteration t	scalar
$R_i^{(t)}$	Euclidean distance $\ \mathbf{x}_i^{(t)} - \mathbf{x}_{\text{best}}^{(t)}\ $	scalar
$\mathcal{U}(a, b)$	Continuous uniform distribution on $[a, b]$	distribution
$\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$	Multivariate Gaussian with mean $\boldsymbol{\mu}$, covariance $\boldsymbol{\Sigma}$	distribution
$[L_j, U_j]$	Box constraint for dimension j	interval
C_f	Cost of one objective-function evaluation	cost unit

Because f is treated as a black box, LUNA has no access to gradient information and must rely solely on point-wise evaluations $f(\mathbf{x})$. This places LUNA in the family of *derivative-free*, *population-based* metaheuristics, alongside PSO, GA, DE, GSA, and WOA.

4 Algorithm Components

LUNA v1 consists of five algorithmic components that together fully specify the algorithm's behavior. This section formalizes each in turn.

4.1 Population Initialization

The initial population $\mathcal{P}^{(0)} = \{\mathbf{x}_1^{(0)}, \mathbf{x}_2^{(0)}, \dots, \mathbf{x}_N^{(0)}\}$ is generated by uniform random sampling within Ω :

$$x_{i,j}^{(0)} \sim \mathcal{U}(L_j, U_j), \quad i \in \{1, \dots, N\}, j \in \{1, \dots, D\}. \quad (2)$$

The corresponding fitness values are computed as $\mathcal{F}_i^{(0)} = f(\mathbf{x}_i^{(0)})$. The best-so-far position is then initialized as

$$\mathbf{x}_{\text{best}}^{(0)} = \mathbf{x}_k^{(0)}, \quad k = \arg \min_{i \in \{1, \dots, N\}} \mathcal{F}_i^{(0)}. \quad (3)$$

Remark 4.1. LUNA v1 does not employ quasi-random sequences (e.g., Sobol, Halton) or opposition-based learning for initialization. This is a known limitation shared with PSO and WOA, but a potential improvement axis for LUNA v2 (cf. Section 10).

4.2 Phase-Angle Dynamics

The phase angle $\alpha^{(t)} \in [0, 2\pi)$ is the principal control variable that schedules the transition from exploration to exploitation. In LUNA v1 it is governed by a *linear* scheduler:

$$\alpha^{(t)} = \frac{t}{T} \cdot 2\pi, \quad t \in \{0, 1, \dots, T-1\}. \quad (4)$$

The transition between phases occurs at $\alpha^{(t)} = \pi$, which corresponds to $t = T/2$. The interpretation is metaphorical: as the lunar phase waxes from new moon ($\alpha = 0$) to full moon ($\alpha = \pi$) the algorithm explores broadly; as it wanes back from full moon to new moon ($\pi \leq \alpha < 2\pi$) the algorithm exploits the best-so-far region.

Definition 4.1 (Phase Function). The phase function $\phi : \{0, \dots, T-1\} \rightarrow \{\text{EXPL}, \text{EXPL}\}$ is defined as

$$\phi(t) = \begin{cases} \text{EXPL} & \text{if } \alpha^{(t)} < \pi, \\ \text{EXPL} & \text{if } \alpha^{(t)} \geq \pi. \end{cases} \quad (5)$$

Remark 4.2. Definition 4.1 implies a *hard, binary* transition. The first half of iterations is purely exploratory; the second half is purely exploitative. There is no overlap, no smooth interpolation, and no stochastic mixing. This is identified as Gap 2 in Section 10.

4.3 Exploration Operator (Gaussian Random Walk)

During the exploration phase ($\phi(t) = \text{EXPL}$), each individual performs an isotropic Gaussian random walk:

$$\mathbf{x}_i^{(t+1)} = \mathbf{x}_i^{(t)} + \boldsymbol{\epsilon}_i^{(t)}, \quad \boldsymbol{\epsilon}_i^{(t)} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_D), \quad (6)$$

with $\sigma = 0.5$ (a static default in LUNA v1). The proposed position is then projected onto Ω by clipping:

$$\tilde{x}_{i,j}^{(t+1)} = \text{clip}(x_{i,j}^{(t)} + \epsilon_{i,j}^{(t)}; L_j, U_j) = \max(L_j, \min(U_j, x_{i,j}^{(t)} + \epsilon_{i,j}^{(t)})). \quad (7)$$

Remark 4.3. Equation (6) is mathematically identical to the mutation operator of the (μ, λ) -Evolution Strategy with isotropic step size σ (Rechenberg, 1973; Schwefel, 1981). The only difference is that LUNA's selection is greedy and per-individual (Section 4.5), whereas classical ES uses comma- or plus-selection at the population level. This equivalence is formalized in Section 8.

4.4 Exploitation Operator (Gravitational Pull)

During the exploitation phase ($\phi(t) = \text{EXPL}$), each individual is attracted toward the best-so-far position $\mathbf{x}_{\text{best}}^{(t)}$ via a simplified gravitational force:

$$\mathbf{x}_i^{(t+1)} = \mathbf{x}_i^{(t)} + \mathbf{F}_i^{(t)}, \quad \mathbf{F}_i^{(t)} = \xi_i^{(t)} \cdot G \cdot \frac{\mathbf{x}_{\text{best}}^{(t)} - \mathbf{x}_i^{(t)}}{R_i^{(t)} + \varepsilon}, \quad (8)$$

where $R_i^{(t)} = \|\mathbf{x}_i^{(t)} - \mathbf{x}_{\text{best}}^{(t)}\|$ is the Euclidean distance to the best, $\xi_i^{(t)} \sim \mathcal{U}(0.5, 1.0)$ is a per-individual stochastic multiplier, $G = 2.5$ is a static gravitational constant, and $\varepsilon = 10^{-3}$ is a numerical safety term that prevents division by zero when $\mathbf{x}_i^{(t)}$ coincides with $\mathbf{x}_{\text{best}}^{(t)}$. The proposed position is again projected onto Ω via clipping (Equation 7).

Remark 4.4. Equation (8) is a special case of the Gravitational Search Algorithm (GSA) force law (Rashedi et al., 2009) in which (i) all masses are set to unity, (ii) only the single best individual acts as the attractor (rather than the population-weighted resultant), and (iii) the gravitational constant is static rather than exponentially decaying. It is also a noisy variant of Central Force Optimization (Formato, 2007).

4.5 Greedy Selection

LUNA v1 uses *per-individual greedy selection*: the proposed position $\tilde{\mathbf{x}}_i^{(t+1)}$ replaces $\mathbf{x}_i^{(t)}$ if and only if it yields a strict improvement in fitness:

$$\mathbf{x}_i^{(t+1)} = \begin{cases} \tilde{\mathbf{x}}_i^{(t+1)} & \text{if } f(\tilde{\mathbf{x}}_i^{(t+1)}) < \mathcal{F}_i^{(t)}, \\ \mathbf{x}_i^{(t)} & \text{otherwise.} \end{cases} \quad (9)$$

The best-so-far position is updated whenever any individual finds a fitness lower than the current incumbent:

$$\mathbf{x}_{\text{best}}^{(t+1)} = \begin{cases} \mathbf{x}_i^{(t+1)} & \text{if } \exists i : \mathcal{F}_i^{(t+1)} < f(\mathbf{x}_{\text{best}}^{(t)}), \\ \mathbf{x}_{\text{best}}^{(t)} & \text{otherwise.} \end{cases} \quad (10)$$

Property 4.1 (Monotone Best-Fitness Decrease). Under the selection rule (9)–(10), the sequence of best-fitness values $\{f(\mathbf{x}_{\text{best}}^{(t)})\}_{t \geq 0}$ is non-increasing:

$$f(\mathbf{x}_{\text{best}}^{(t+1)}) \leq f(\mathbf{x}_{\text{best}}^{(t)}), \quad \forall t \geq 0. \quad (11)$$

Proof. By Equation (10), $\mathbf{x}_{\text{best}}^{(t+1)}$ is set to $\mathbf{x}_i^{(t+1)}$ only if $f(\mathbf{x}_i^{(t+1)}) < f(\mathbf{x}_{\text{best}}^{(t)})$; otherwise it retains its previous value. Hence $f(\mathbf{x}_{\text{best}}^{(t+1)}) \leq f(\mathbf{x}_{\text{best}}^{(t)})$ in both cases. By induction the sequence is non-increasing. $\square$

5 Formal Pseudocode

Algorithm 1 consolidates the five components of Section 4 into a single, self-contained specification. The pseudocode uses the notation of Section 2 and is suitable for direct implementation.

Algorithm 1: LUNA v1 — Lunar-inspired Update & Navigation Algorithm

Input: Objective $f : \mathbb{R}^D \rightarrow \mathbb{R}$; bounds $\{[L_j, U_j]\}_{j=1}^D$; population size N ; max iterations T ; gravitational constant G ; step-size σ ; safety ε .

Output: Best-so-far solution $\mathbf{x}_{\text{best}}$ and its fitness $\mathcal{F}_{\text{best}}$.

```

/* -- Initialization -- */
1 for  $i \leftarrow 1$  to  $N$  do
2   for  $j \leftarrow 1$  to  $D$  do
3      $x_{i,j}^{(0)} \sim \mathcal{U}(L_j, U_j)$ ;
4   end
5    $\mathcal{F}_i^{(0)} \leftarrow f(\mathbf{x}_i^{(0)})$ ;
6 end
7  $k \leftarrow \arg \min_i \mathcal{F}_i^{(0)}$ ;
8  $\mathbf{x}_{\text{best}} \leftarrow \mathbf{x}_k^{(0)}$ ;  $\mathcal{F}_{\text{best}} \leftarrow \mathcal{F}_k^{(0)}$ ;

/* -- Main Loop -- */
9 for  $t \leftarrow 0$  to  $T - 1$  do
10   $\alpha \leftarrow (t/T) \cdot 2\pi$ ;
11  for  $i \leftarrow 1$  to  $N$  do
12    if  $\alpha < \pi$  then
13      /* Exploration: Gaussian random walk */
14       $\epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_D)$ ;
15       $\tilde{\mathbf{x}} \leftarrow \mathbf{x}_i^{(t)} + \epsilon$ ;
16    else
17      /* Exploitation: gravitational pull toward best */
18       $R \leftarrow \|\mathbf{x}_i^{(t)} - \mathbf{x}_{\text{best}}\|$ ;
19       $\xi \sim \mathcal{U}(0.5, 1.0)$ ;
20       $\tilde{\mathbf{x}} \leftarrow \mathbf{x}_i^{(t)} + \xi \cdot G \cdot (\mathbf{x}_{\text{best}} - \mathbf{x}_i^{(t)}) / (R + \varepsilon)$ ;
21    end
22    /* Project onto feasible region */
23    for  $j \leftarrow 1$  to  $D$  do
24       $\tilde{x}_j \leftarrow \max(L_j, \min(U_j, \tilde{x}_j))$ ;
25    end
26     $\tilde{\mathcal{F}} \leftarrow f(\tilde{\mathbf{x}})$ ;
27    if  $\tilde{\mathcal{F}} < \mathcal{F}_i^{(t)}$  then
28       $\mathbf{x}_i^{(t+1)} \leftarrow \tilde{\mathbf{x}}$ ;  $\mathcal{F}_i^{(t+1)} \leftarrow \tilde{\mathcal{F}}$ ;
29      if  $\tilde{\mathcal{F}} < \mathcal{F}_{\text{best}}$  then
30         $\mathbf{x}_{\text{best}} \leftarrow \tilde{\mathbf{x}}$ ;  $\mathcal{F}_{\text{best}} \leftarrow \tilde{\mathcal{F}}$ ;
31      end
32    else
33       $\mathbf{x}_i^{(t+1)} \leftarrow \mathbf{x}_i^{(t)}$ ;  $\mathcal{F}_i^{(t+1)} \leftarrow \mathcal{F}_i^{(t)}$ ;
34    end
35  end
36 end
37 return  $\mathbf{x}_{\text{best}}, \mathcal{F}_{\text{best}}$ ;

```

6 Parameter Specification

Table 2 enumerates every parameter of LUNA v1, its symbol, default value in the released prototype, parameter class (static vs. dynamic), and a brief remark on tuning strategy. Parameters that are flagged as *static* in LUNA v1 are precisely those targeted by the v2 enhancements in Section 10.

Table 2: Parameter specification for LUNA v1.

Symbol	Name	Default	Class	Remark
N	Population size	30	Static	Standard metaheuristic tuning; $N \in [20, 100]$ typical.
T	Max iterations	500–15000	Static	Should be set so that $N \cdot T \approx 10^4$ – 10^5 for fair benchmarking.
D	Problem dimension	(problem)	–	Set by the application instance.
$[L_j, U_j]$	Box constraints	$[-10, 10]$	Static	LUNA v1 uses symmetric bounds; can be generalized per dimension.
G	Gravitational constant	2.5	Static	<i>Gap 1 target</i> : should be time-varying in v2 (e.g., $G(t) = G_0 \cdot e^{-\alpha t/T}$ as in GSA).
σ	Random-walk step size	0.5	Static	<i>Gap 1 target</i> : should be adaptive in v2 (e.g., sigma-decay or 1/5-success rule).
ε	Safety constant	10^{-3}	Static	Numerical safeguard; not a tuning parameter.
ξ	Stochastic multiplier	$\mathcal{U}(0.5, 1.0)$	Dynamic	Per-individual, per-iteration random draw; already adaptive.
$\alpha^{(t)}$	Phase angle	$\frac{t}{T}2\pi$	Linear	<i>Gap 2 target</i> : hard π -threshold should become smooth in v2.

Critical Remark on Reproducibility

The LUNA v1 source code does not fix the pseudo-random number generator seed, making experimental results non-reproducible. For any benchmark comparison, the seed must be fixed (e.g., `numpy.random.seed(42)`) and reported alongside results.

7 Computational Complexity Analysis

This section derives the worst-case time and space complexity of LUNA v1 in terms of the population size N , problem dimension D , iteration budget T , and the cost of a single objective-function evaluation C_f . The analysis assumes that sampling from $\mathcal{U}$ and $\mathcal{N}$ costs $\mathcal{O}(1)$ per scalar draw.

7.1 Time Complexity

Initialization (line 1–5 of Algorithm 1). Sampling $N \cdot D$ uniform scalars costs $\mathcal{O}(ND)$. Computing N fitness evaluations costs $\mathcal{O}(NC_f)$. The argmin over N fitness values costs $\mathcal{O}(N)$.

Total initialization cost:

$$T_{\text{init}} = \mathcal{O}(ND + NC_f + N) = \mathcal{O}(N(D + C_f)). \quad (12)$$

Main loop (line 7–end of Algorithm 1). Per individual per iteration, the costs are:

- Phase-angle update: $\mathcal{O}(1)$.
- Exploration branch: sampling $\epsilon \in \mathbb{R}^D$ costs $\mathcal{O}(D)$; vector addition costs $\mathcal{O}(D)$.
- Exploitation branch: distance computation $\mathcal{O}(D)$; scalar multiply + vector add $\mathcal{O}(D)$.
- Clipping: $\mathcal{O}(D)$.
- Fitness evaluation: $\mathcal{O}(C_f)$.
- Selection and best-update: $\mathcal{O}(1)$.

Hence per-iteration per-individual cost is $\mathcal{O}(D + C_f)$. Summed over N individuals and T iterations:

$$T_{\text{main}} = \mathcal{O}(T \cdot N \cdot (D + C_f)). \quad (13)$$

Total time complexity. Combining initialization and main loop, and observing that $T \geq 1$:

$$T_{\text{total}} = T_{\text{init}} + T_{\text{main}} = \mathcal{O}(N(D + C_f) + TN(D + C_f)) = \mathcal{O}(TN(D + C_f)). \quad (14)$$

Proposition 7.1 (Time Complexity of LUNA v1). The worst-case time complexity of LUNA v1 is

$$\mathcal{O}(T \cdot N \cdot (D + C_f)), \quad (15)$$

where T is the iteration count, N the population size, D the problem dimension, and C_f the cost of one objective-function evaluation.

7.2 Space Complexity

LUNA v1 stores: (i) the population matrix of size $N \times D$; (ii) the fitness vector of size N ; (iii) the best-so-far position $\mathbf{x}_{\text{best}} \in \mathbb{R}^D$; (iv) a small number of scalar accumulators. Hence

$$S_{\text{total}} = \mathcal{O}(ND + N + D) = \mathcal{O}(ND). \quad (16)$$

7.3 Comparison with Peer Algorithms

Table 3 compares LUNA’s asymptotic complexity with five canonical peers. The comparison confirms that LUNA lies in the same complexity class as established population-based metaheuristics: none of the algorithms is asymptotically faster or slower than the others, and the dominant cost in all cases is the per-iteration population-wide fitness evaluation $\mathcal{O}(TNC_f)$.

Remark 7.1. GSA has a $\mathcal{O}(N^2)$ factor in its main loop because each individual computes a pairwise force against every other individual. LUNA v1 avoids this by attracting each individual only to the single best-so-far position, reducing the per-iteration auxiliary cost to $\mathcal{O}(ND)$. This is a computational advantage of LUNA over GSA when N is large, at the cost of losing population-level information aggregation.

8 Mathematical Comparison with Related Algorithms

To position LUNA v1 in the metaheuristic landscape and to anticipate reviewer scrutiny (especially the metaphor-critique raised by Sörensen, 2015, and Camacho-Villalón, 2018), this section performs a direct operator-level comparison with five canonical algorithms: PSO, GA, DE, GSA, and WOA. Table 4 tabulates the comparison; the discussion that follows interprets it.

Table 3: Asymptotic complexity of LUNA v1 vs. canonical peers. All algorithms share the same dominant term $\mathcal{O}(TN(D + C_f))$; the per-individual auxiliary cost differs slightly.

Algorithm	Time complexity	Space complexity
LUNA v1 (this work)	$\mathcal{O}(TN(D + C_f))$	$\mathcal{O}(ND)$
PSO (Kennedy & Eberhart, 1995)	$\mathcal{O}(TN(D + C_f))$	$\mathcal{O}(ND)$
GA (Holland, 1992)	$\mathcal{O}(TN(D + C_f))$	$\mathcal{O}(ND)$
DE (Storn & Price, 1997)	$\mathcal{O}(TN(D + C_f))$	$\mathcal{O}(ND)$
GSA (Rashedi et al., 2009)	$\mathcal{O}(TN^2(D + C_f))$	$\mathcal{O}(ND)$
WOA (Mirjalili & Lewis, 2016)	$\mathcal{O}(TN(D + C_f))$	$\mathcal{O}(ND)$

Table 4: Operator-level mathematical comparison of LUNA v1 with PSO, GA, DE, GSA, and WOA. “Equiv.” denotes mathematical equivalence to an existing operator; “Adapt.” denotes an adaptive (non-static) variant.

Component	LUNA v1	PSO	GA	DE	GSA	WOA
Position update	$\mathbf{x} + \epsilon$ or $\mathbf{x} + \xi G \frac{\mathbf{x}_{\text{best}} - \mathbf{x}}{R}$	$\mathbf{x} + \mathbf{v}$; $\mathbf{v} \leftarrow w\mathbf{v} + c_1 r_1 \Delta \mathbf{p} + c_2 r_2 \Delta \mathbf{g}$	$\mathbf{x}_c = \text{cross}(\mathbf{x}_a, \mathbf{x}_b) + \text{mutate}$	$\mathbf{x} + F(\mathbf{x}_a - \mathbf{x}_b) + F(\mathbf{x}_c - \mathbf{x}_d)$	$\mathbf{x} + \sum_j \mathbf{F}_{ij}$	Encircling / spiral / random
Exploration mechanism	Isotropic Gaussian walk, $\sigma = 0.5$ static	Velocity drift + cognitive term	Bit-flip / Gaussian mutation	Difference-vector mutation	Inertia from pairwise forces	Random shrink encircling
Exploitation mechanism	Gravitational pull toward $\mathbf{x}_{\text{best}}$	Social term (pull toward $\mathbf{g}$)	Elitism + crossover	Crossover with target vector	Pull toward mass-weighted centroid	Spiral around best
Phase transition	Hard switch at $\alpha = \pi$	Smooth (via $w(t)$ decay)	None explicit	None explicit	Smooth (via $G(t)$ decay)	Hard switch at $p = 0.5$
Personal-best memory	None	Yes ($\mathbf{p}_i$)	Implicit (via elitism)	None	None	None
Selection	Per-individual greedy	Replace if better	Generational / steady-state	Greedy (target vs. trial)	Per-individual greedy	Per-individual greedy
Random params	$\xi \sim \mathcal{U}(0.5, 1)$, $\epsilon \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$	w, c_1, c_2 (often linearly decaying)	p_c, p_m	F, CR	G_0, α (decaying)	a, A, C, p, l (decaying)

Three observations follow directly from Table 4.

First, LUNA v1’s exploration operator (Equation 6) is mathematically a special case of (μ, λ) -ES mutation with isotropic step size $\sigma = 0.5$. The metaphor of “lunar waxing” plays no role in the equation itself; it only schedules when the operator is invoked. A reviewer familiar with Sörensen (2015) will note that this falls under the “rebranding” critique unless a v2 enhancement justifies the lunar framing.

Second, LUNA v1’s exploitation operator (Equation 8) is a special case of GSA’s force law with all masses set to unity, the attractor reduced from the population centroid to the single best-so-far, and G kept static. Compared with DE’s difference-vector mutation $\mathbf{x} + F(\mathbf{x}_a - \mathbf{x}_b)$, LUNA’s exploitation differs only by the normalization $\frac{1}{R+\varepsilon}$ and the stochastic multiplier ξ .

Third, the phase-transition mechanism of LUNA v1 is a hard binary switch at $\alpha = \pi$, which is structurally parallel to WOA’s hard switch at $p = 0.5$. By contrast, PSO and GSA achieve exploration–exploitation balance through smoothly time-varying parameters ($w(t)$ and $G(t)$, respectively), which is now considered best practice in the metaheuristic community. The hard switch is identified as Gap 2 in Section 10.

Implication for the SLR

The comparison above indicates that LUNA v1, as it currently stands, is a *composition* of two well-known operators—ES mutation and GSA force law—scheduled by a WOA-style hard switch. To claim genuine algorithmic novelty in the upcoming SLR and benchmark paper, at least one of the three enhancements proposed in Section 10 must be incorporated into LUNA v2 and empirically validated against the canonical baselines listed in Table 4.

9 Convergence Properties (Preliminary)

A complete convergence proof of LUNA v1 is beyond the scope of this document and, given the algorithm’s current design, is unlikely to be obtainable in a strong form (cf. the parallel difficulty of proving convergence for PSO and GSA). Nevertheless, three properties can be stated.

Property 9.1 (Bounded Best-Fitness Sequence). If f is bounded below on Ω , then $\{f(\mathbf{x}_{\text{best}}^{(t)})\}_{t \geq 0}$ converges, since by Property 4.1 it is non-increasing and bounded below.

Remark 9.1. Property 9.1 guarantees only that the best-fitness sequence converges to some limit $f^* \geq f(\mathbf{x}^*)$; it does not guarantee that the limit equals the global optimum $f(\mathbf{x}^*)$. In particular, LUNA v1 can stagnate in a local optimum during the exploitation phase, especially because the gravitational pull (Equation 8) is unidirectional toward $\mathbf{x}_{\text{best}}^{(t)}$ with no escape mechanism.

Property 9.2 (Phase-1 Diffusion). During the exploration phase ($t < T/2$), if the clipping operation is inactive (i.e., $\mathbf{x}_i^{(t)}$ lies in the interior of Ω), the expected position of individual i is preserved:

$$\mathbb{E}[\mathbf{x}_i^{(t+1)} \mid \mathbf{x}_i^{(t)}] = \mathbf{x}_i^{(t)}, \quad (17)$$

since $\mathbb{E}[\boldsymbol{\epsilon}_i^{(t)}] = \mathbf{0}$. The per-coordinate variance grows linearly in t : $\text{Var}(x_{i,j}^{(t)}) = \text{Var}(x_{i,j}^{(0)}) + t\sigma^2$ (before any selection event). Selection introduces a downward bias on fitness but does not change the zero-mean nature of the perturbation.

Remark 9.2. Property 9.2 suggests that, in expectation, the exploration phase performs an unbiased random walk. This is theoretically attractive for global search but, combined with greedy selection, can lead to premature convergence: once an individual reaches a locally good region, the greedy rule preserves it indefinitely, and the random walk has no incentive to leave. This risk is amplified by the static step size $\sigma = 0.5$, which is not adapted to the landscape’s local geometry. This is part of Gap 1 in Section 10.

10 Identified Novelty Gaps and LUNA v2 Roadmap

The formalization in Sections 4–8 reveals three concrete gaps that limit LUNA v1’s theoretical contribution relative to existing algorithms. Each gap is paired below with a formalized v2 enhancement. Together, these enhancements would constitute a defensible algorithmic novelty claim.

10.1 Gap 1: Static Parameter Problem

Gap 1 — Static Parameters

LUNA v1 uses a static gravitational constant $G = 2.5$ and a static step size $\sigma = 0.5$ throughout the entire run. By contrast, GSA uses an exponentially decaying $G(t) = G_0 e^{-\alpha t/T}$, PSO uses a linearly decaying inertia weight $w(t)$, and modern ES use the 1/5-success-rule or CMA-ES self-adaptation. Static parameters are widely recognized as a weakness because they cannot balance global search (favored by large G , large σ) and local refinement (favored by small G , small σ) across the run.

Proposed v2 enhancement. Replace the static G and σ by time-varying, lunar-cycle-inspired modulations:

$$G^{(t)} = G_0[1 + \kappa \sin(\omega t)], \quad (18)$$

$$\sigma^{(t)} = \sigma_0(1 - t/T) + \sigma_{\min}, \quad (19)$$

Equation (18) introduces a sinusoidal modulation with n_{cyc} lunar cycles per run (e.g., $n_{\text{cyc}} = 3$), making the gravitational constant oscillate around G_0 with amplitude κG_0 , where $\omega = 2\pi n_{\text{cyc}}/T$. This is a genuine use of the lunar-cycle metaphor: the gravitational pull genuinely waxes and wanes, in contrast to the metaphor-only use in v1. Equation (19) applies a linear decay from σ_0 to $\sigma_{\min}$, mirroring standard practice in PSO and GSA.

10.2 Gap 2: Hard Phase Transition

Gap 2 — Hard Phase Transition

LUNA v1 transitions from exploration to exploitation via a binary switch at $\alpha = \pi$ (Definition 4.1). This means the algorithm cannot perform exploration in the second half of the run, even if the landscape demands it. By contrast, PSO and GSA use continuous interpolation, and recent algorithms (e.g., HHO, SMA) use probabilistic or fuzzy transitions.

Proposed v2 enhancement. Replace the binary phase function with a smooth probabilistic transition:

$$\Pr[\phi(t) = \text{EXPL}] = \frac{1}{1 + \exp\left(-\beta\left(\frac{t}{T} - \frac{1}{2}\right)\right)}, \quad (20)$$

so that the algorithm randomly samples, per individual per iteration, which operator to apply. The parameter β controls the sharpness of the transition: $\beta \rightarrow \infty$ recovers the hard v1 switch; $\beta = 0$ gives a uniform 50/50 mixture throughout the run. A recommended default is $\beta = 8$.

10.3 Gap 3: Absence of Personal-Best Memory

Gap 3 — No Personal-Best Memory

LUNA v1 tracks only the global best $\mathbf{x}_{\text{best}}^{(t)}$. PSO's success is widely attributed to its dual memory structure: each individual also remembers its own historical best $\mathbf{p}_i$, which preserves useful sub-region information even when $\mathbf{x}_{\text{best}}$ has moved elsewhere. The lack of personal-best memory in LUNA v1 may contribute to premature convergence (Remark 9.2).

Proposed v2 enhancement. Introduce per-individual personal-best memory $\mathbf{p}_i^{(t)}$ and modify the exploitation operator:

$$\mathbf{x}_i^{(t+1)} = \mathbf{x}_i^{(t)} + \xi G^{(t)}(\lambda_g \mathbf{d}_g + \lambda_p \mathbf{d}_p), \quad (21)$$

where $\mathbf{d}_g = \frac{\mathbf{x}_{\text{best}}^{(t)} - \mathbf{x}_i^{(t)}}{R_g + \varepsilon}$ and $\mathbf{d}_p = \frac{\mathbf{p}_i^{(t)} - \mathbf{x}_i^{(t)}}{R_p + \varepsilon}$ are unit-norm direction vectors toward the global and personal best respectively, $\lambda_g, \lambda_p \geq 0$ are global and personal-pull weights (e.g., $\lambda_g = \lambda_p = 0.5$), and R_g, R_p are the corresponding Euclidean distances. The personal best is updated as in PSO: $\mathbf{p}_i^{(t+1)} = \arg \min(f(\mathbf{x}_i^{(t+1)}), f(\mathbf{p}_i^{(t)}))$.

Combined v2 Novelty Claim

If the three enhancements above are integrated into LUNA v2, the resulting algorithm is no longer a mere composition of ES mutation and GSA force law. It becomes (i) a sinusoidally modulated metaheuristic whose parameter dynamics genuinely encode the lunar metaphor, (ii) a probabilistic-phase-switch algorithm distinct from WOA's binary switch, and (iii) a dual-memory algorithm that combines the strengths of GSA (gravitational pull) and PSO (personal best). This combination is, to the best of our knowledge, novel in the metaheuristic literature and constitutes a defensible contribution claim.

11 Closing Remarks and Next Steps

This document has provided a complete mathematical formalization of LUNA v1, including its five algorithmic components (Section 4), full pseudocode (Section 5), parameter specification (Section 6), and a worst-case complexity analysis demonstrating that LUNA runs in $\mathcal{O}(TN(D + C_f))$ time and $\mathcal{O}(ND)$ space (Section 7). A side-by-side comparison with PSO, GA, DE, GSA, and WOA (Section 8) revealed that LUNA v1 is, mathematically, a composition of an Evolution-Strategy-style Gaussian mutation and a simplified GSA gravitational pull, scheduled by a WOA-style hard phase switch. Three concrete novelty gaps were identified and paired with formalized v2 enhancements (Section 10).

The formalization serves as the theoretical foundation for two downstream research deliverables. The first is a Systematic Literature Review (SLR) of physics- and astronomy-inspired metaheuristics, in which LUNA v2 must be taxonomically positioned relative to its true mechanistic cousins (GSA, MVO, BHA, TFA, BB-BC, CFO, GbSA). The SLR's four research questions are: (RQ1) how are exploration operators formulated in physics-inspired metaheuristics; (RQ2) how are phase-transition strategies designed; (RQ3) to what extent do the metaphors of these algorithms genuinely manifest in their mathematics, versus serving as a rebranding of classical operators; and (RQ4) what mechanistic research gaps can be addressed by a new algorithm. The LUNA v2 enhancements proposed in Section 10 provide direct answers to RQ4.

The second downstream deliverable is the LUNA v2 benchmark paper, which must compare LUNA v2 against the canonical baselines of Table 4 on a standard benchmark suite (e.g., CEC

2017 or CEC 2022) over 30 independent runs, with statistical validation via the Wilcoxon rank-sum test at $\alpha = 0.05$. The formalization in this document specifies exactly what must be implemented and what parameters must be tuned, ensuring reproducibility.

The immediate next deliverable in the research pipeline is the SLR protocol document (PRISMA 2020 compliant), including the PICOC-framed research questions, the Boolean search strings for Scopus, Web of Science, and IEEE Xplore, the inclusion and exclusion criteria, and the quality-assessment checklist (JBI or MMAT). That protocol will operationalize the four research questions stated above and produce the literature corpus against which LUNA v2's novelty claim will be empirically grounded.